

DOING PHYSICS WITH PYTHON

CONDUCTION OF THERMAL ENERGY

MODELLING ENERGY LOSSES FROM A ROOM

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tp001.py

Thermal conduction: energy lost from a room through a brick wall slab and a glass window.

tp002.py

Thermal conduction: energy lost from a room through a brick wall with a wood insulation fibre.

tp003.py

Thermal conduction: energy lost from a room through a double brick wall slab with a wood fibre insulation filling the cavity between the bricks.

INTRODUCTION

The transfer of heat is very important in industry, in the home and in biological systems. At times it is desirable to obtain the maximum heat transfer, at other times to reduce it. It is essential that the thermal properties of materials from the lowest to the highest temperatures are known. Then systems can be designed to meet the requirements of maximum or minimum transfer of energy. For example, what is the purpose of the large cooling towers in a power station? How can we keep our home warm in winter? How does our body try to keep us cool during a hot summer day? On a very hot day you pick up two bars lying on the ground in the sun, one is made of wood, the other metal, what would you feel and why? In answering these questions, you have to consider, what produces a transfer of energy, and what causes a change in temperature. Your answers may include the terms hot, cold, temperature, heat, energy, energy transfer, power, energy flux, and energy flux density but what do these terms mean? Can you clearly differentiate the concepts associated with each of these terms?

You can gain a deeper and better understanding of these concepts, by modelling the transfer of energy through a slab

(material) by running the simulations with a wide range of input parameters.

Heat is a misleading term

Heat is the term used to describe the transfer of energy due to a temperature difference.

A better alternative is to simply use the term **energy** and so on most occasions the word energy will be used and not the term heat. The **rate of energy transfer** dQ/dt across a surface is referred to as the **power** or **energy flux**. The **energy flux density** J is a very useful term because it is independent of the cross-sectional area A of the slab. The **energy flux density** J is defined as

$$(1) \quad J = \frac{1}{A} \frac{dQ}{dt}$$

We will model the one-dimensional heat transfer through a slab. The model can be used to compute both the temperature and the transfer of energy as functions of time and position.

The rate of energy transfer dQ/dt through a slab is dependent upon the thermal conductivity k of the material, the cross-sectional area A of the slab and the temperature gradient dT/dx as described by the differential equation

$$(2) \quad \frac{dQ}{dt} = -k A \frac{dT}{dx}$$

Using equations (1) and (2), the energy flux density J is

$$(3) \quad J = -k \frac{dT}{dx}$$

When energy is transferred to or from an object its temperature may change. The change in temperature dT depends upon the object's mass m , its specific heat capacity c and the net amount of energy Q_{net} gained or lost by the object.

The change in temperature dT is given by

$$(4) \quad dT = \frac{Q_{net}}{m c}$$

and the time rate of change of temperature is

$$(5) \quad \frac{dT}{dt} = \frac{1}{m c} \frac{dQ_{net}}{dt}$$

We can consider the width of the slab to be made up of many elements of length dx . The density of an element is ρ and its volume is $V = A dx$. Then the mass m of an element is

$$(6) \quad m = \rho A dx$$

Combining equations (1), (4) and (5), the time rate of change of the temperature of an element is

$$(7) \quad \frac{dT}{dt} = \frac{1}{\rho c A dx} \frac{dQ_{net}}{dt} = \frac{1}{\rho c dx} J_{net} \quad J_{net} = \frac{1}{A} \frac{dQ_{net}}{dt}$$

where the net energy flux density J_{net} is given by

$$(8) \quad J_{net} = J_{cond} + J_{int} + J_{env}$$

J_{cond} is the contribution to the energy flux density due to the conduction of energy through the slab, J_{int} is the internal heating of the slab (for example, an electric heating element) and J_{env} is the due to the transfer of energy to and from an element to the surrounding environment (for example, by thermal radiation or convection).

We can find the temperature T and energy flux density J_{net} as functions of time t and position x through the slab by solving the two differential equations (3) and (7) using a **finite difference approach**

$$(3) \quad J_{cond} = -k \frac{dT}{dx}$$

$$(7) \quad \frac{dT}{dt} = \frac{1}{\rho c} \frac{dJ_{net}}{dx} \quad J_{net} = J_{cond} + J_{int} + J_{env}$$

For convenience, to simplify the notation, we will use J to represent the energy flux density due to conduction $J \equiv J_{cond}$.

Consider a slab of width L divided into N elements and the width of each element is $\Delta x = L / N$. It is assumed all the properties of an element are uniform throughout the volume of the element. For the n^{th} element ($n = 1, 2, \dots, N$), it is characterised by its thermal conductivity $k(n)$, specific heat capacity $c(n)$ and its density $\rho(n)$. The time dependent variables are its temperature $T(t, n)$ and energy flux densities $J_{\text{int}}(t, n)$ and $J_{\text{env}}(t, n)$. All these variables are measured at the mid-point of the element $x(n) = \Delta x / 2 + (n - 1) \Delta x$.

Energy is transferred across the interface between elements by the process of conduction. The energy flux density into the n^{th} element is $J(t, n)$ at position $x_j(n) = (n - 1) \Delta x$ and the energy flux density existing the element is $J(t, n+1)$ at position $x_j(n+1) = x_j(n) + \Delta x$. Figure 1 shows the parameters for the element with index n at time t .

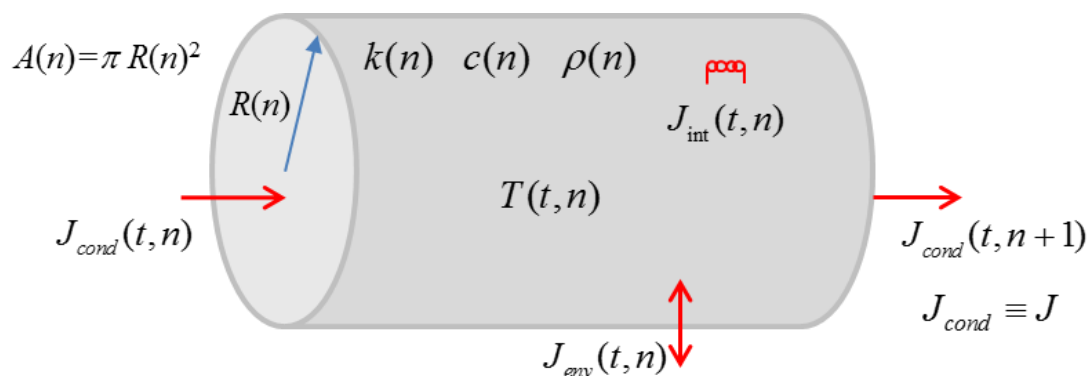


Fig. 1. Parameters for the element with index n at time t .

The differential equation (3) for the energy flux density can be expressed as a finite difference equation for the n^{th} element at time t as

$$(8) \quad J(t, x_n) = -\frac{k_n}{\Delta x} \{T(t, x_{n+1}) - T(t, x_n)\}$$

The differential equation (7) for the temperature can be expressed as a finite difference equation for the n^{th} element at time $t + \Delta t$ as

$$(9) \quad T(t + \Delta t, x_n) = T(t, x_n) + \frac{\Delta t}{\rho c dx} \{J(t, x_{J_{n-1}}) - J(t, x_{J_n}) + J_{\text{int}}(t, x_n) + J_{\text{env}}(t, x_n)\}$$

We can solve this pair of finite difference equations (8) and (9) given the initial values at time $t = 0$ and the boundary conditions imposed on the ends of the slab.

For stability of the numerical method, we have the condition

$$\frac{\Delta t}{\rho c dx} < 1$$

The smaller the value dx for the increment in x then the smaller the time step dt in the simulation.

For composite materials made of more than one material, it may not be possible to use actual values for the heat capacity and density as they set the time scale in solving the differential equations. The steady-state temperature profile along a slab and the energy flux density only depend upon the width and thermal conductivity of the materials making up the composite material, therefore, values of the heat capacity and density can be chosen to give an appropriate time step.

SIMULATIONS

Consider a room of a house where the outside temperature is $5\text{ }^{\circ}\text{C}$ and the inside room temperature is $25\text{ }^{\circ}\text{C}$. We can model the loss of energy from the room to the outside through a brick wall and through a glass window by conduction. The cross-section area of the slab for the brick and glass is 1.0 m^2 . The initial temperature for each element of the slab is $5\text{ }^{\circ}\text{C}$ except the end element which is set to the room temperature, $25\text{ }^{\circ}\text{C}$.

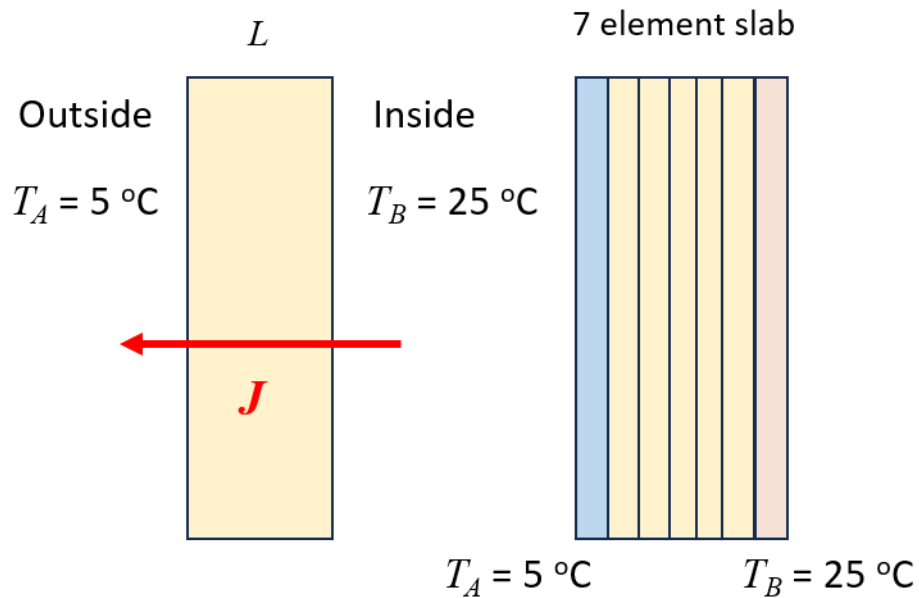


Fig. 1. Energy flux density J is transmitted from room to the outside of the house because of the temperature difference due to the boundary conditions $T_A = 5\text{ }^{\circ}\text{C}$ (first element of slab) and $T_B = 25\text{ }^{\circ}\text{C}$ (end element of slab).

Simulation 1 Energy flux density through brick and glass

tp001.py

The inputs are:

```
#%% INPUTS
```

```
# Input flagS: building brick 1 or glass window 2
```

```
flagS = 1 # <<<<<
```

```
# Building brick
```

```
if flagS == 1:
```

```
    nx = 121 # number of spatial elements (101)
```

```
    nt = 30000 # number of time steps (50000)
```

```
    L = 0.08 # width
```

```
    k1 = 1.33 # thermal conductivity W/(m.K)
```

```
    rho1 = 1600 # density kg/m**3
```

```
    c1 = 900 # specific heat capacity J/(kg.K)
```

```
# Glass window
```

```
if flagS == 2:
```

```
    nx = 121 # number of spatial elements (101)
```

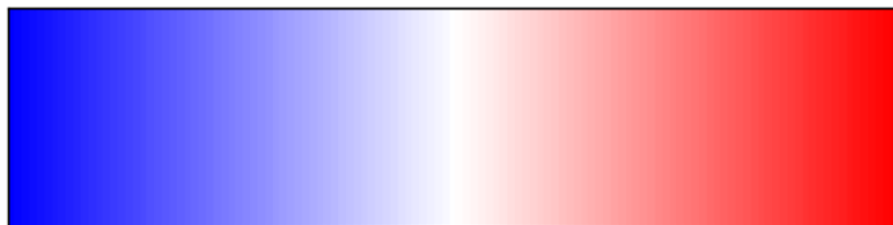
```
    nt = 30000 # number of time steps (50000)
```

```
    L = 3e-3 # width
```

```
    k1 = 0.8 # thermal conductivity W/(m.K)
```

```
    rho1 = 2600 # density kg/m**3
```

```
    c1 = 840 # specific heat capacity J/(kg.K)
```



$T_A = 5\text{ }^{\circ}\text{C}$

$T_B = 25\text{ }^{\circ}\text{C}$

Fig 1.1 Pcolor plot of the steady-state temperature profile across the slab.

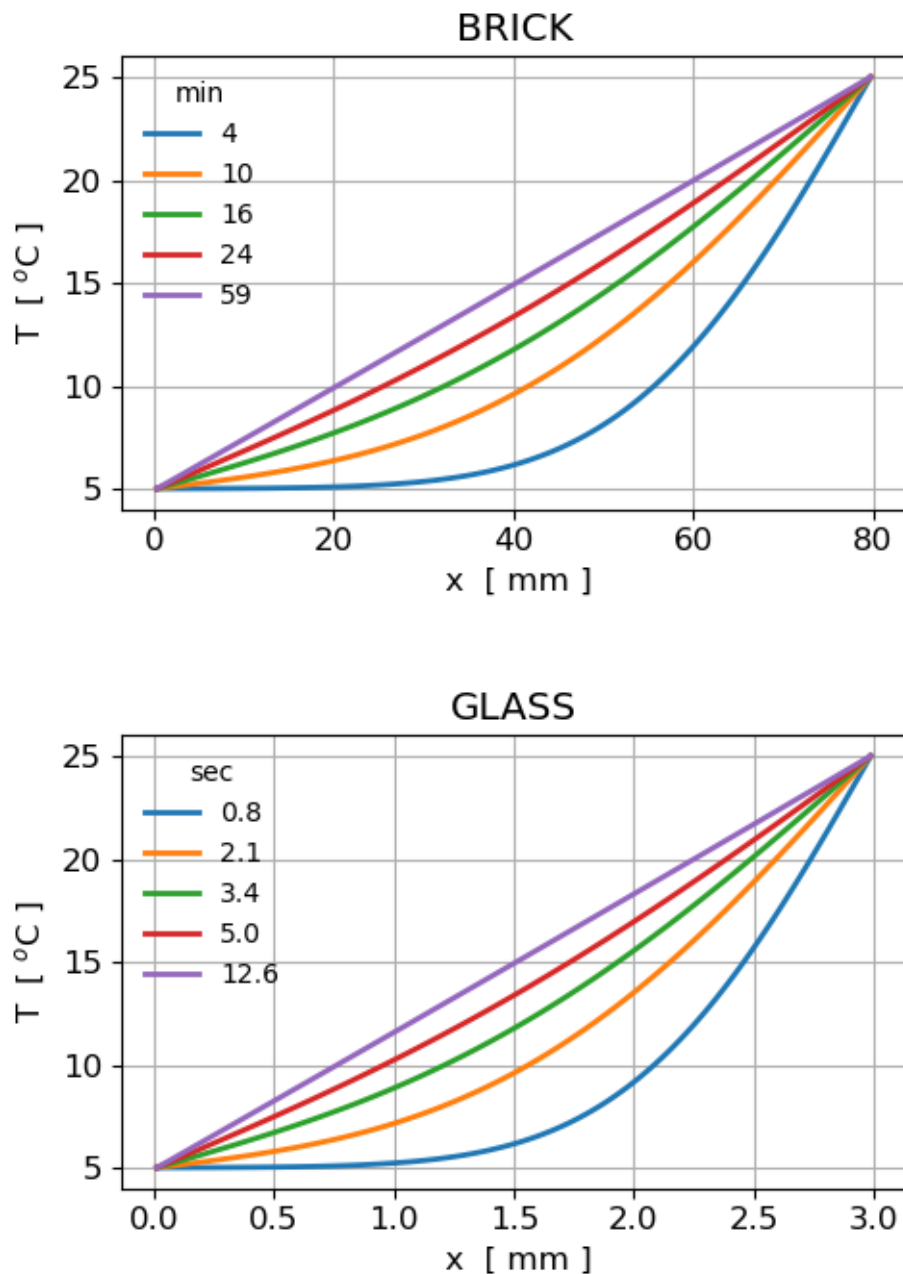


Fig. 1.2. Temperature distribution through the slab. The steady-state temperature profile is reached in about 1 seconds for the glass but about 1 hour for the brick.

Steady-state temperature gradients:

$$\text{brick} = -2.5 \times 10^2 \text{ }^{\circ}\text{C} \cdot \text{m}^{-1}$$

$$\text{glass} = -6.7 \times 10^3 \text{ }^{\circ}\text{C} \cdot \text{m}^{-1}$$

Glass temperature gradient is ~ 27 x greater than the brick

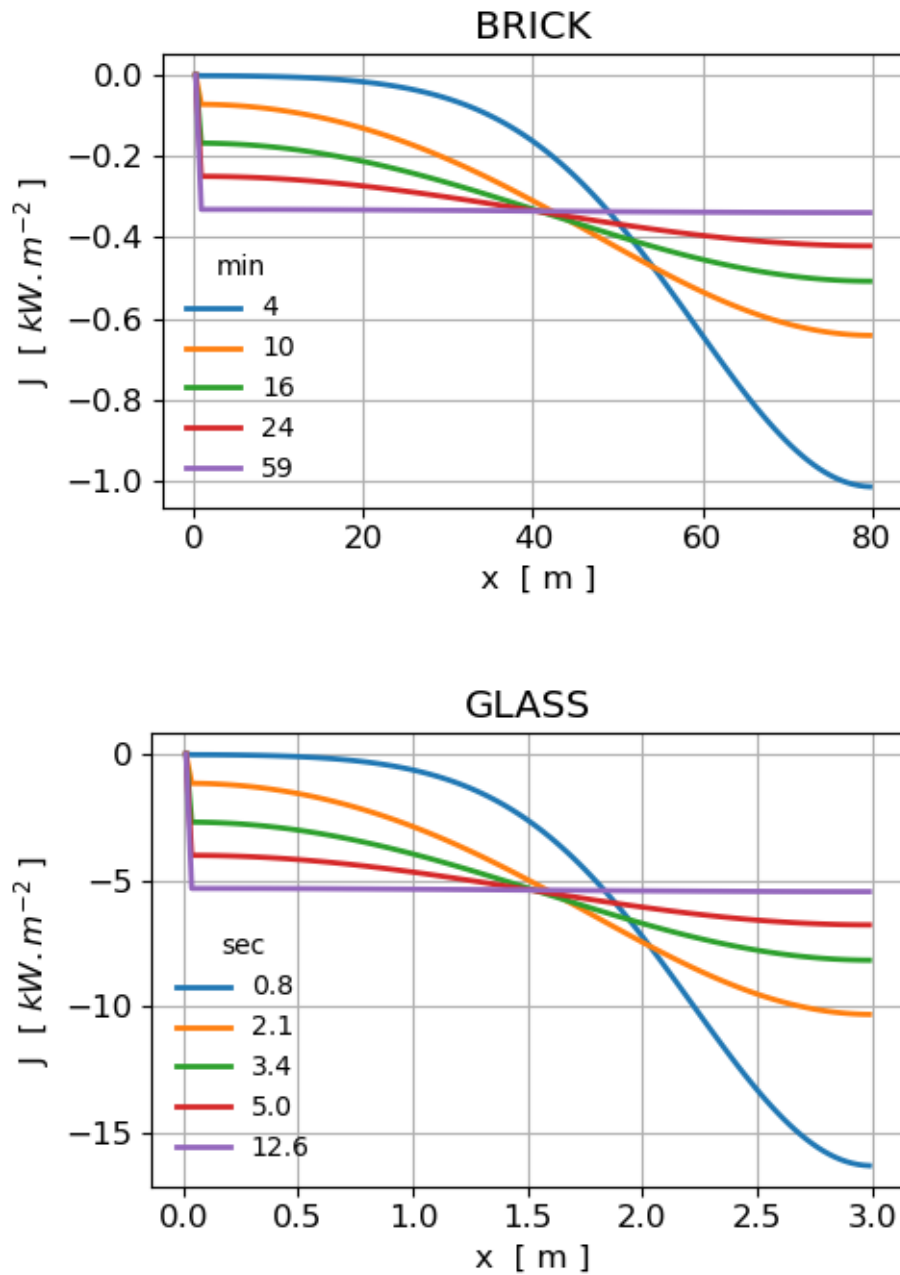


Fig. 1.3. Energy flux density through the brick and glass slabs. A constant energy flux density through the glass takes about 10 seconds and about 1 hour for the brick.

Steady-state energy flux density J

$$J_{brick} = -0.34 \text{ kW.m}^{-2} \quad J_{glass} = -5.4 \text{ kW.m}^{-2}$$

The rate of energy lost from the room to the outside is **18x** more through the glass than the brick.

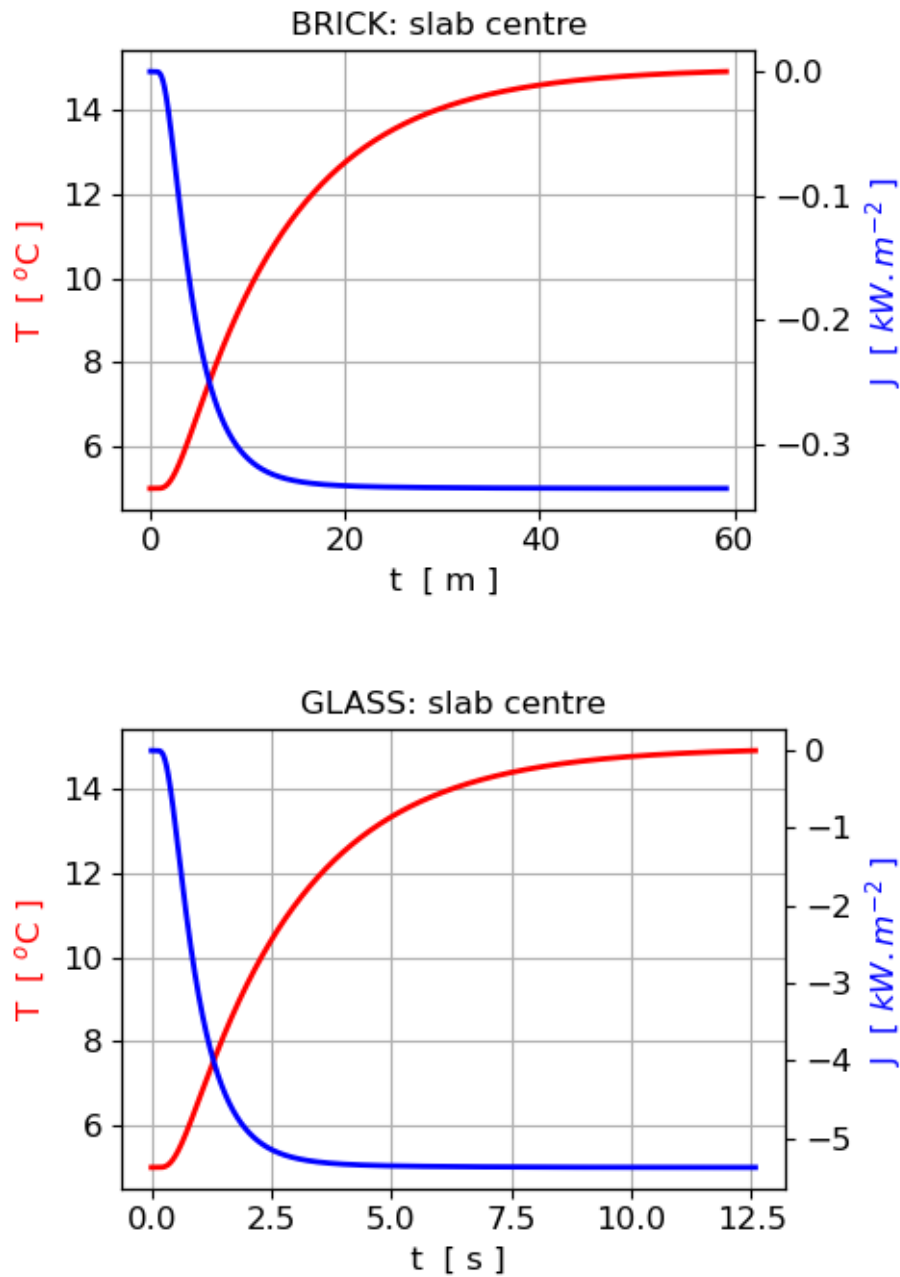


Fig. 1.4. The time evolution of the temperature and energy flux density at the centre of the slab. The steady-state is reached for the glass much quicker than for the brick.

Energy is transferred through the slabs from the hot reservoir to the cold reservoir. Initially a large amount of energy is transferred through the slab in the direction of decreasing temperature, heating the slab which results in the rise in

temperature along its length. Finally, a steady-state situation is established with a linear variation in temperature through the slab and a constant energy flux density through it.

The significance of the minus sign ($J < 0$) is that the energy transfer is in the direction of decreasing temperature, i.e., heat always flows spontaneously from the location of higher temperature to locations of lower temperature.

For an energy efficient home, the lower the rate of energy lost from the interior to the external the better. We can model a double brick wall ($L = 8.0 \text{ mm} \rightarrow 16 \text{ mm}$) and a better glass window for insulation ($L = 3 \text{ mm} \rightarrow 10 \text{ mm}$) to achieve better insulation to reduce heat losses.

Brick: $L = 0.08 \text{ m}$ $J_{brick} = -0.34 \text{ kW.m}^{-2}$

$$L = 0.16 \text{ m} \quad J_{brick} = -0.17 \text{ kW.m}^{-2}$$

The double brick slab halves the energy lost.

Glass: $L = 3.0 \text{ mm}$ $J_{glass} = -5.44 \text{ kW.m}^{-2}$

$$L = 10 \text{ mm} \quad J_{glass} = -1.63 \text{ kW.m}^{-2}$$

The thick glass slab reduces the energy lost by a factor of 3 but the energy loss is still much much greater than the loss of energy through the brick.

Simulation 2 Conduction through a composite slab

Brick and Wood Fibre Insulation

tp002.py

Energy flow calculations often involve composite walls that are composed of more than one material. For insulation purposes, these composite walls have a layer of a material with low thermal conductivity.

If the slab is only brick of thickness 75 mm thick, then the temperature difference of 20 °C gives a steady-state energy flux density of **360 W.m⁻²**. So, considerable energy losses occur through the brick walls of the room.

To improve the insulation of a house, an insulating fibre can be inserted into wall cavities. We can model this by considering a composite slab of brick with thickness 75 mm and a wood fibre insulating material of thickness 75 mm. The parameters for wood fibre are:

thermal conductivity $k = 0.038 \text{ W.m}^{-1}.\text{K}^{-1}$

density $\rho = 200 \text{ kg.m}^{-3}$

specific heat capacity $c = 2100 \text{ J.kg}^{-1}.\text{K}^{-1}$

The slab's temperature will evolve with time from its initial temperature to a steady-state situation when the energy flux density is uniform through the width of the slab (the rate of energy transfer through the two materials are equal) where

$$(2.1) \quad J_{cond} = -k_1 \frac{\Delta T_1}{\Delta x_1} = -k_2 \frac{\Delta T_2}{\Delta x_2}$$

From equation 2.1 it is easy to show that the theoretical temperature at the junction between the brick and fibre is

$$(2.2) \quad T_{junction} = \frac{T_A + (k_1 / k_2) T_B}{1 + (k_1 / k_2)} = 5.6 \text{ } ^\circ C$$

The theoretical value agrees with the numerical prediction. The larger drop in temperature occurs across the fibre because the fibre has a much lower the conductivity than the brick.

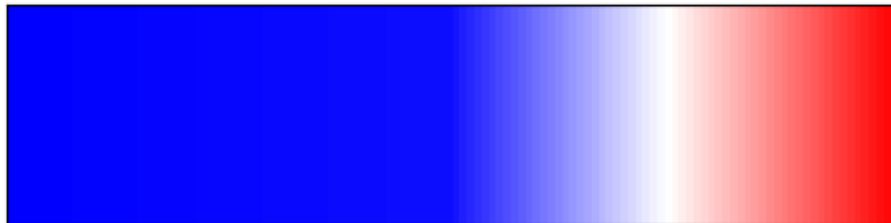


Fig. 2.1. Temperature across brick-fibre slab.

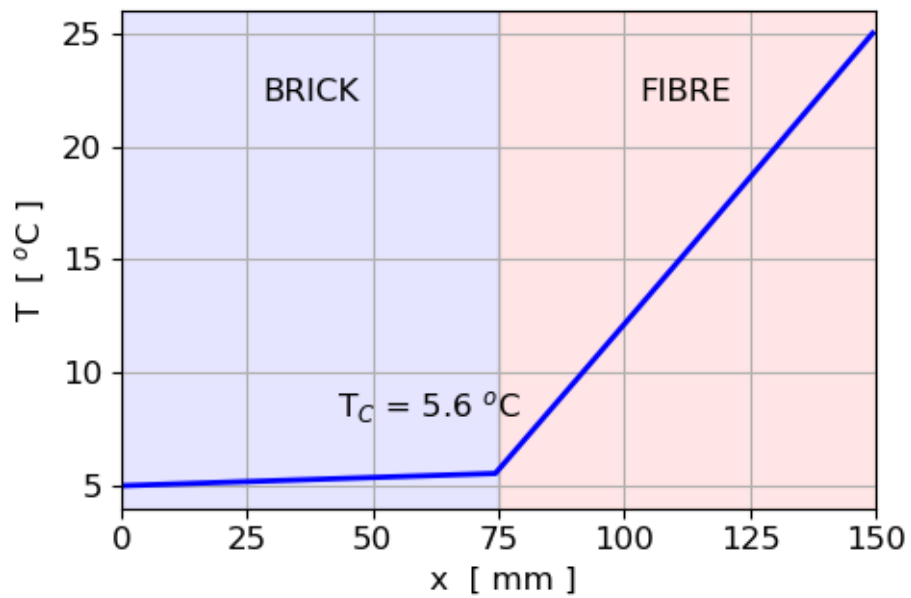


Fig. 2.2. Steady-state temperature distribution through the brick-fibre slab.

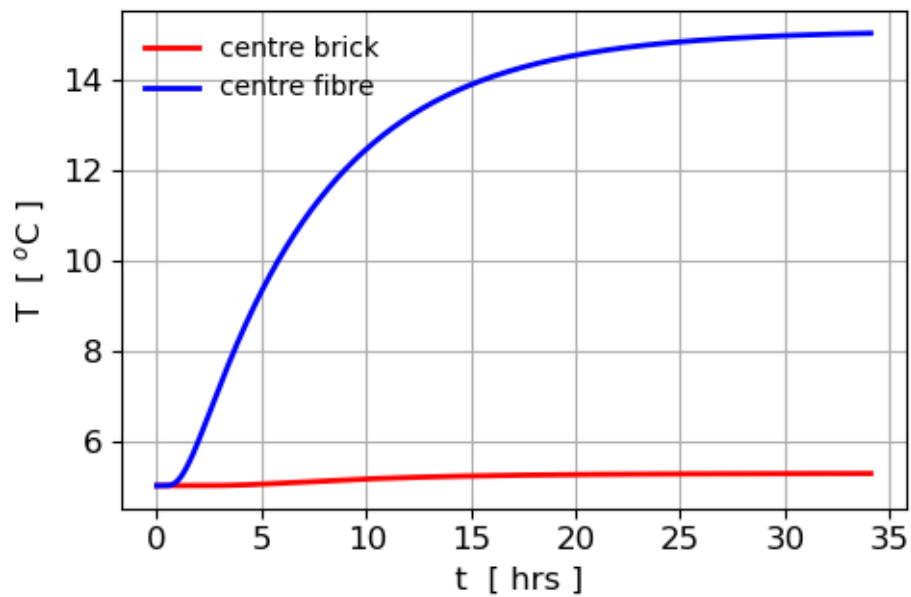


Fig. 2.3. The temperature evolution at the centre of the brick and centre of the fibre. Steady-state values are obtained only after many hours, especially for the brick.

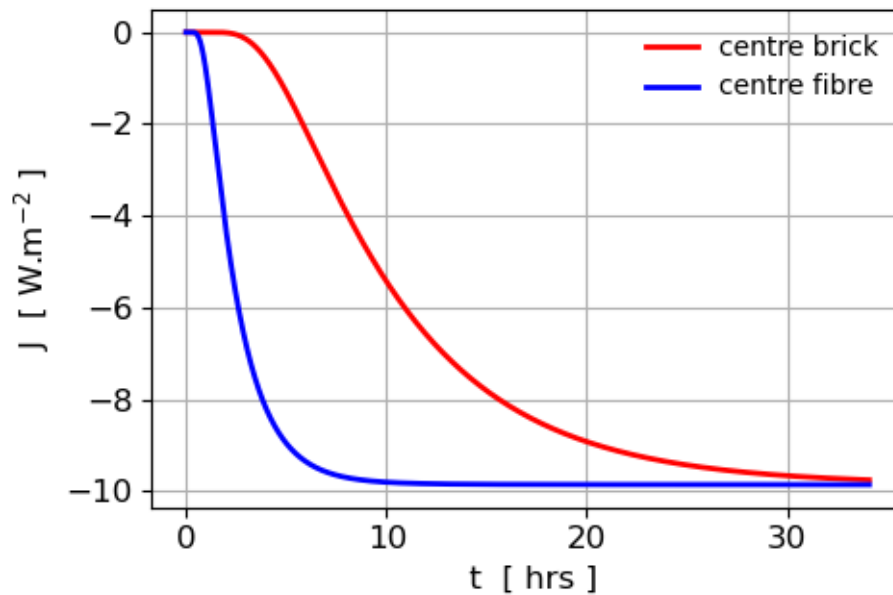


Fig. 2. 4. Time evolution of the energy flux density through the brick and the fibre. It takes much more time for the steady-state energy flux density through the brick to be reached compared with the fibre.

The loss of energy through the brick is **360 W.m⁻²** where after a layer of wood fibre insulation is added, there is a significant reduction in the energy lost through the insulated wall. The energy flux density drops to **10 W.m⁻²**.

Simulation 3 Filled cavity double brick wall

We will consider a specific example of a composite wall of double brick with the cavity between the brick walls filled with a wooden insulation fibre. The width of the filled cavity is 20 mm and the width of the slab is 170 mm.

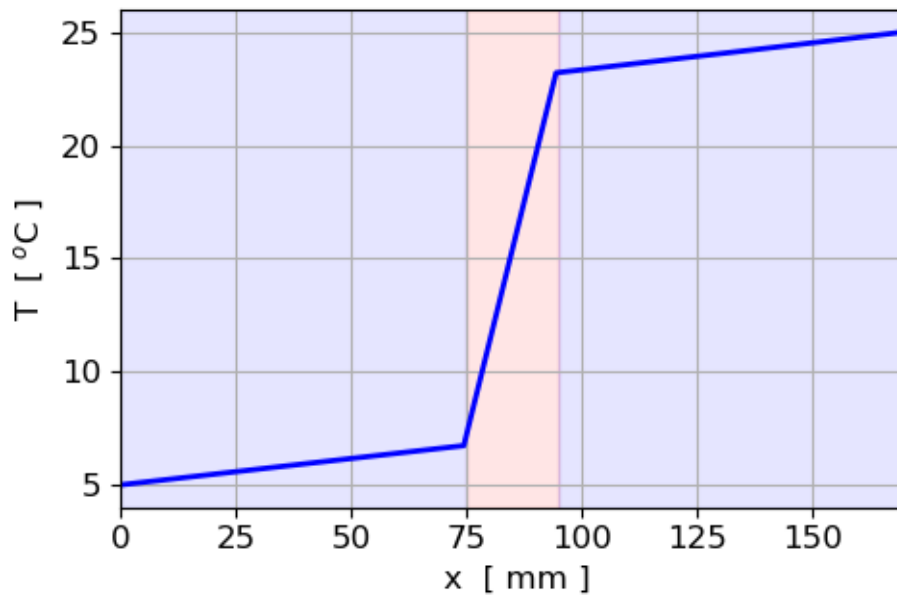


Fig 3.1. The temperature distribution through the wall segment. The large temperature gradient occurs across the wood fibre filled cavity.

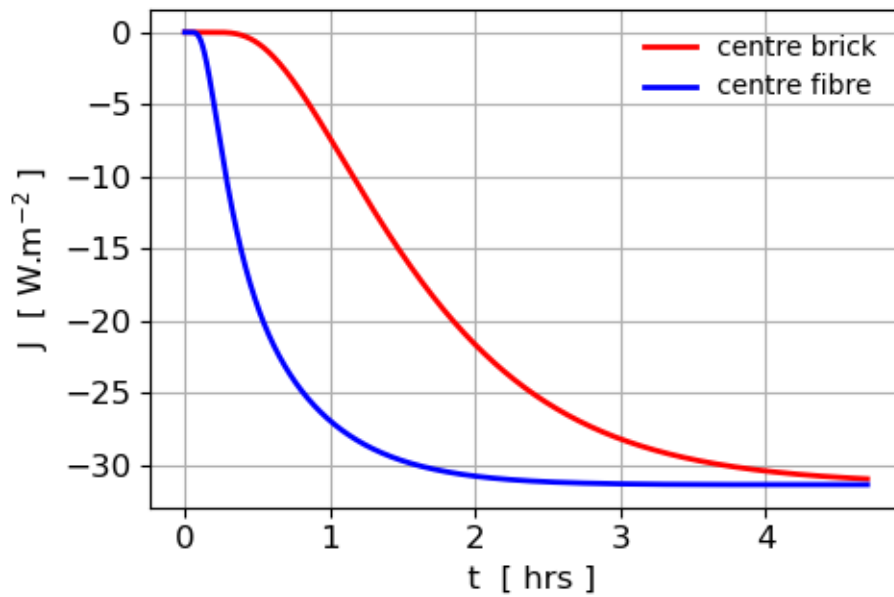


Fig. 3.2. Time evolution of the energy flux density. The steady-state energy flux density is -32 W.m^{-2} .

The energy lost through a double brick wall (150 mm width) is 180 W.m^{-2} . By adding the wood fibre insulation (20 mm width) into the cavity reduces the energy loss to 32 W.m^{-2} .